CODE:

```
import math


# Initialize board
board = [" " for _ in range(9)]


def print_board(board):
    for row in [board[i*3:(i+1)*3] for i in range(3)]:
        print("| " + " | ".join(row) + " |")


def available_moves(board):
    return [i for i, spot in enumerate(board) if spot == " "]


def winner(board):
    win_combos = [
        [0,1,2], [3,4,5], [6,7,8],  # rows
        [0,3,6], [1,4,7], [2,5,8],  # columns
        [0,4,8], [2,4,6]            # diagonals
    ]
    for combo in win_combos:
        if board[combo[0]] == board[combo[1]] == board[combo[2]] != " ":
            return board[combo[0]]
    return None


def minimax(board, depth, is_maximizing):
    if winner(board) == "X":
        return 1
    elif winner(board) == "O":
```

```
        return -1
    elif " " not in board:
        return 0

    if is_maximizing:
        best_score = -math.inf
        for move in available_moves(board):
            board[move] = "X"
            score = minimax(board, depth+1, False)
            board[move] = " "
            best_score = max(score, best_score)
        return best_score
    else:
        best_score = math.inf
        for move in available_moves(board):
            board[move] = "O"
            score = minimax(board, depth+1, True)
            board[move] = " "
            best_score = min(score, best_score)
        return best_score

def best_move(board):
    best_score = -math.inf
    move = None
    for i in available_moves(board):
        board[i] = "X"
        score = minimax(board, 0, False)
        board[i] = " "
```

```
        if score > best_score:
            best_score = score
            move = i
    return move

def play_game():
    print("Tic Tac Toe - You are O, AI is X")
    print_board(board)

    while True:
        # Human move
        human_move = int(input("Enter your move (0-8): "))
        if board[human_move] == " ":
            board[human_move] = "O"
        else:
            print("Invalid move. Try again.")
            continue

        if winner(board) or " " not in board:
            break

        # AI move
        ai_move = best_move(board)
        board[ai_move] = "X"

        print_board(board)

        if winner(board) or " " not in board:
```

```
        break


    print_board(board)
    if winner(board) == "X":
        print("AI wins!")
    elif winner(board) == "O":
        print("You win!")
    else:
        print("It's a draw!")


play_game()
```

OUTPUT:

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/AI/EXP 13.py ======
Tic Tac Toe - You are O, AI is X
|   |   |   |
|   |   |   |
|   |   |   |
Enter your move (0-8): 0
| O |   |   |
|   | X |   |
|   |   |   |
Enter your move (0-8): 1
| O | O | X |
|   | X |   |
|   |   |   |
Enter your move (0-8): 7
| O | O | X |
| X | X |   |
|   | O |   |
Enter your move (0-8): 6
| O | O | X |
| X | X | X |
| O | O |   |
| O | O | X |
| X | X | X |
| O | O |   |
AI wins!
```